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Roll No

MI-5001-CBGS

B.E. V Semester

Examination, June 2020

Choice Based Grading System (CBGS)

Surface Mining

Time : Three Hours

Maximum Marks : 70

Note : i) Attempt any five questions.

ii) All questions carry equal marks.

1. a) Describe productivity calculation of Dragline, BWE, Rope shovel.
b) Describe Internal and external Box Cut with diagram.
2. a) Explain placer and solution mining.
b) What type of mineral deposits are amenable to surface mining? Discuss with suitable examples.
3. a) Describe the main constructional features of a scraper with neat sketch.
b) Explain the walking mechanism and walking operation of a dragline with suitable sketch.
4. a) Explain how the height, width and slope of a bench in an open pit mine is determined.
b) Discuss the conditions under which rotary blast hole drilling machine are preferred and its constructional features.

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5. a) Discuss the various activities of open pit mine development in brief and the various site preparation activities required.
b) Discuss the various types of Scrapers. Explain machine operation in brief.
6. a) Describe various factors which govern Bench Parameters of an opencast mine?
b) i) Natural stripping ratio
ii) Average Stripping ratio
iii) Peak stripping ratio
7. a) Describe safety features of modern off highway dumpers.
b) Write down the applicable conditions of Conveyor Transport System in Mines.

OR

What are the various types of off-highway dumpers used in opencast mines? Describe them.

8. a) What is meant by synchronization of shovel dumper combination? Write down the conditions under which Bottom Discharge Dumpers are preferred?
b) Write a note on high angle belt conveyor and pipe conveyor.

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